



Fall 2023

Exam Date: 12.11.2023

Exam Time 60 min.

MEP 221s: Fluid Mechanics and Turbomachinery
MEP231: Fluid Dynamics
Midterm Examination

The Exam Consists of **three** Questions in **Three** Pages

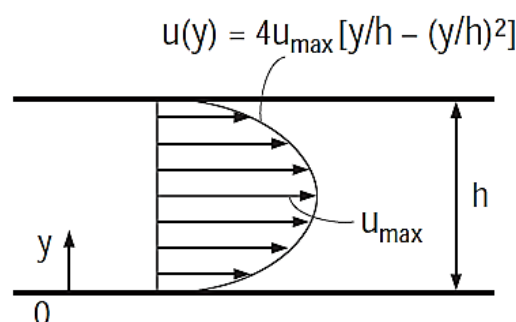
Total Marks: 25 Marks

Student Name:

ID:

Question (1): (8 points)

Consider the laminar flow of a Newtonian fluid of viscosity μ between two parallel plates. The flow is one-dimensional, and the velocity profile is given as $u(y) = 4u_{\max} [(y/h) - (y/h)^2]$, where y is the vertical coordinate from the bottom surface, h is the distance between the two plates, and $u_{\max}$ is the maximum flow velocity that occurs at the midplane.



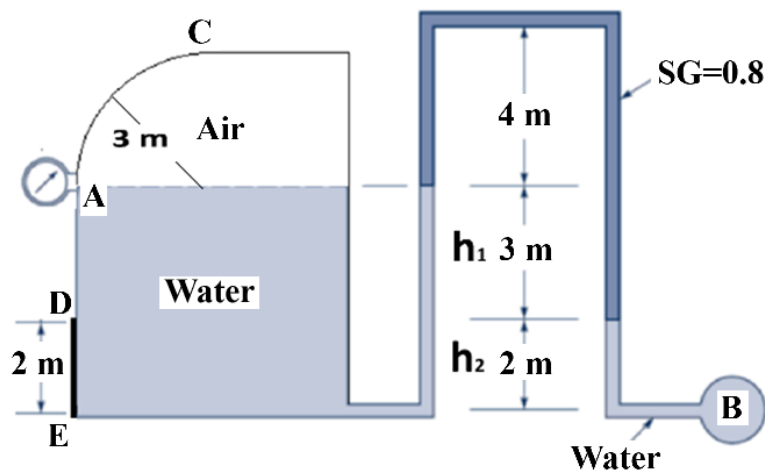
Develop a relation for the drag force exerted on both plates by the fluid in the flow direction per unit area of the plates.

MEP 221s: Fluid Mechanics and Turbomachinery
MEP231: Fluid Dynamics

Question (2): (9 points)

A closed tank containing water is connected to an inverted U-tube manometer. A rectangular plate **DE** of 2 m height and 1 m width is placed at the bottom of the tank and a 1 m-long curved gate **AC** is located at the top left side of the tank as shown in the Figure. The liquid in the top part of the inverted U-tube manometer has a specific gravity of 0.8, and the remaining is filled with water. The upper part of the tank is filled with air. If the pressure at **B** is 205 kPa absolute and the atmospheric pressure is 1 bar. Determine:

- (a) The pressure gauge reading at **A**. (2points)
- (b) The magnitude, direction, and location of the resultant hydrostatic force acting on the curved gate **AC**. (2 points)
- (c) The magnitude, direction, and location of the resultant hydrostatic force acting on the plate **DE** (using the pressure prism method only). (5 points)

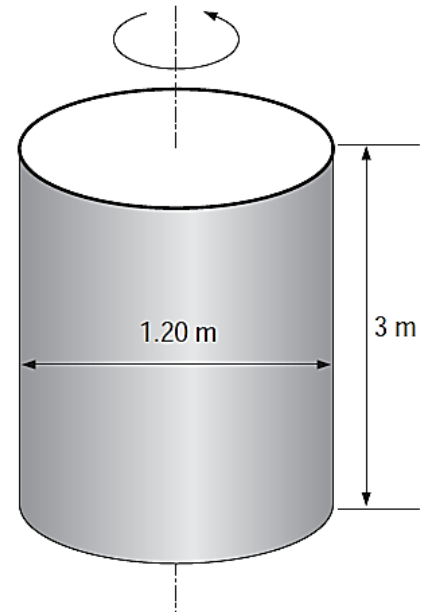


MEP 221s: Fluid Mechanics and Turbomachinery
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Question (3): (8 points)

A 1.2-m-diameter, 3-m-high sealed vertical cylinder is completely filled with gasoline whose density is 740 kg/m^3 . The tank is now rotated about its vertical axis at a rate of 70 rpm. **Determine**

- a- the difference between the pressures at the centers of the bottom and top surfaces of the cylinder. **(4 points)**
- b- the difference between the pressures at the center and the edge of the bottom surface. **(4 points)**



Examination Committee:

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